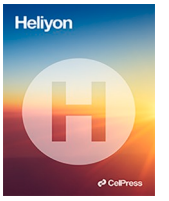
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Research article



Continuous monitoring of body temperature for objective detection of health and safety risks in construction sites: An analysis of the accuracy and comfort of off-the-shelf wearable sensors

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ABSTRACT

Recent studies have shown the potential of wearable sensors for objective detection of health and safety risks in construction workers through their collected physiological data. Body temperature, as the focus of the current study, is one of the most important physiological parameters that can help to detect various health and safety risks such as heat stress, physical fatigue, and infectious diseases. This study aims to assess the applicability and performance of off-the-shelf wearable sensor devices to monitor workers' body temperature in construction sites by evaluating the accuracy of temperature measurements as well as the comfort of the devices.

A total of nine off-the-shelf wearable sensor devices available on the market were initially trialed in the laboratory, and three devices were shortlisted considering a set of selection criteria for further assessment. Over three weeks, the shortlisted wearable sensors were tested on 26 workers in two large construction sites in Australia. The reliability/validity of the selected wearable sensors in measuring body temperature was investigated using Bland-Altman analysis. Human factors were also investigated in terms of the comfort of the devices, their impact on workers' performance, and the acceptability of being worn for an extended period (i.e., 8 h or more). It was found that all selected devices measured body temperature with a bias of less than one indicating a slight difference in measurements compared to the reference hospital-grade thermometers. Two devices out of the three were also comfortable. The achieved results indicate that it is feasible to develop a continuous temperature monitoring platform using off-the-shelf wearable sensors to detect a range of significant health and safety risks in construction sites objectively. Considering the rapid advancements in manufacturing wearable sensors, future research can adopt a similar approach to include the newly introduced off-the-shelf temperature sensors and select the most appropriate device.

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1. Introduction

The construction industry has one of the highest rates of work-related injuries and accidents and has the third-highest fatality rate of any industry with an average of 3.1 fatalities per 100,000 workers each year. This rate was 15% higher than the 5-year average for this industry [1]. These alarming statistics indicate that the industry has reached saturation with respect to the traditional health and safety improvement strategies and new safety innovations are needed to save the lives of many workers.

Wearable sensors provide an efficient solution to improve health and safety in construction sites. These inexpensive, small, lightweight, and smart sensor devices can be used to measure various safety and health performance metrics in construction [2]. Using wearable sensors, a wide range of physiological parameters such as body temperature, heart rate, breath rate, and brain signals can be monitored in real-time to detect potential safety hazards and monitor a worker's health [3]. Among these physiological parameters, body temperature is one of the most important indicators of significant safety and health risks [4]. Body temperature can aid in detecting the most significant safety risks such as heat stress, physical fatigue, and workers' physical states as well as health issues such as infectious diseases [5–10].

The past studies have verified the importance of body temperature in detecting physical fatigue [5]. Body temperature often indicates the thermoregulation of the body that results from physical activities [11]. When workers perform intense physical activities, metabolic heats leave the body via increased skin temperature [12]. Therefore, it can be used to detect the level of physical fatigue. Body temperature has also been proven to be an indicator of heat stress [10,13]. When heat is produced in the body due to environmental heat, the mechanism of human thermoregulation helps the heat dissipation from the skin surface to the surrounding environment to keep the basic core body temperature at approximately 37 °C [14,15]. When heat is induced in the body, the skin vessels expand, and blood flow in the skin increases, which will thereby increase the skin temperature [16]. Body temperature has also been widely used to detect infectious diseases such as COVID-19 [17–19]. In this sense, several studies have used body temperature to monitor various safety and health risks.

For monitoring body temperature, we can measure the temperature of the skin or core. Core body temperature refers to the temperature of the body's internal organs and its accurate measurement involves an invasive procedure of swallowing temperature pills. Therefore, it is a person's skin temperature that is commonly measured in body sites such as the mouth, ear, and armpit. In the current research, body temperature refers to skin temperature. Traditionally, body temperature is measured using non-contact infrared thermometers [20,21]. However, for detecting safety risks such as heat stress and physical fatigue, there is a need to have regular and continuous temperature measurements at short time intervals. This would not be possible using traditional temperature measurement devices since they will interrupt the work. Therefore, there is a need to embed infrared temperature sensors in a chest strap or ear clip to measure body temperature continuously [5,22,23]. Considering the working conditions in construction sites, an efficient way to provide such continuous temperature measurement for construction workers is by using wearable devices that utilize sensors to capture the temperature in real time [2,24]. Recently, there have been new advancements in manufacturing flexible wearable temperature sensors to better monitor and determine the heat exchanges occurring between the inner body tissues and the external environment [25–29]. The fast developments in thin film manufacturing technology are expected to provide solutions for replacing bulky and rigid electronics, that could easily and fully integrate onto nonplanar surfaces, particularly on the human body. Specifically, the recent advancements in printing functional nanomaterials for wearable temperature sensors on low-cost polymeric substrates can enhance the attractiveness of cost-effective portable and wearable sensing applications [25,26].

By reviewing the literature, it is revealed that there are a very limited number of studies in which wearable sensors have been used for continuous monitoring of body temperature to detect health and safety risks such as physical fatigue, heat stress, and infectious diseases. One of the main reasons could be the difficulty of continuous measurement of body temperature in construction sites due to the specific nature of the jobs. Firstly, to use wearable sensors in construction sites, they should not interfere with the work and should be comfortable to be worn for an extended period (i.e., $\geq$8 h). In addition, it is important to use sensor devices that can accurately measure body temperature while it is affected by several external factors such as workers' movements due to intense physical activities as well as ambient conditions. Past studies which aim to detect safety and health risks using continuous monitoring of body temperature are reviewed in the following.

For the risk of physical fatigue, there is one study conducted in the construction sector to assess the risk of physical fatigue using continuous monitoring of body temperature. Aryal et al. [5] used a construction safety helmet, fitted with four non-contact infrared temperature sensors to monitor the body temperature. They showed that the features extracted from the temperature sensor achieved an accuracy of 79.4% in detecting physical fatigue. However, it is difficult to use their custom-built sensor on a real construction site since it is not a non-intrusive device. In addition, measuring forehead temperature using an infrared sensor may not be accurate [25,26]. Lastly, their study was conducted in a lab with a simulated task done under room temperature. Unlike a lab-based simulated experimental task, there is no control over the factors that may affect body temperature in a construction site. These factors include workers' intense physical activities and their movements which will affect the fit of the device and the accuracy of temperature measurements as well as changes in ambient conditions that affects body temperature.

For the risk of heat stress, there are four previous studies verified the importance of body temperature in heat strain assessment. Pham et al. [30] used a temperature sensor to monitor the temperature of the upper arm. Although the device seems to be comfortable to be used on a construction site, the reported average skin temperature was 32.48 $\pm$ 0.33 °C. They tried to estimate body temperature based on upper arm measurements. However, the upper arm is not a site of the body recommended by reference clinical guides such as the Royal Australian College of General Practitioners (RACGP) [31] to measure body temperature and that's the reason the body temperatures reported by Pham et al. were significantly lower than normal body temperature. In the field of heat stress and physical

demand, Jebelli et al. [8] also conducted a study in which a wristband (E4 by Empatica Inc.) was used to study construction workers' physical demands under environmental heat. They showed that higher variations of an outdoor worker's body temperature as well as an increasing tendency of body temperature can exhibit the combined effects of internal body factors and external environmental factors on physical demands. Although they used a comfortable device, they also measured the temperature of the wrist which can't provide an accurate estimation of body temperature. The wrist is not a site of the body that can represent body temperature [31] and that's the reason their reported temperatures were much lower than the normal range of body temperature (i.e., mainly between 29 and 33 °C). In another study, Shakerian et al. [10] and Ojha et al. [32] used the same wristband device (E4 by Empatica Inc) to assess occupational risk of heat stress at construction. They extracted features from body temperature to assess the risk of heat strain [10,32]. Their study may also suffer from the same limitations stated before for Jebelli et al. [8] since they measured the temperature of a wrist that can't represent actual body temperature. They also conducted their tests in a laboratory setting.

Lastly, for infectious diseases, there is one past study in the construction domain that attempted to continuously measure body temperature to detect symptoms of COVID-19 [33]. They used a simulated construction experiment and measured the forehead temperature continuously using a helmet with a non-contact infrared temperature sensor. However, the comfort of the device and the practicality of using the device in real construction sites were not discussed. In addition, the study was not conducted in a real construction environment to consider the impact of physical activity which may affect the body temperature. The researchers used a walking task, while in a construction site, workers perform several types of activities with different intensities. The intense physical activities will significantly affect the accuracy of measurements due to the movements of the sensor and the fit of the device as well as the effect of physical exertion on the body temperature.

By investigating the previous studies which aim to measure body temperature continuously for objective detection of health and safety risks, it is revealed that they have some limitations.

- Past studies measured the temperature of the forehead, upper arm, or wrist which are not sites of the body recommended by reference clinical guides to measure body temperature. The inner ear (tympanic), underarm (axilla), and oral are the sites of the body that can represent the body temperature [31]. Several studies have criticized the use of the upper arm or wrist for temperature monitoring since their measured temperatures highly fluctuate and are far from the core temperature [34–36]. Also measuring forehead temperature using infrared sensors may not be accurate [25,26].
- The comfort of the devices, their effect on workers' performance, and their acceptability to be worn for an extended period in a construction site (e.g., 8 h or more) were not investigated in the past studies.
- None of the past studies collected data from a real construction site to assess the accuracy and performance of the devices under real working conditions.

This research aims to address the above limitations and find an efficient way for continuous monitoring of body temperature in construction sites that will pave the way toward objective detection of health and safety risks. The research is the first in construction and one of the first studies conducted in a real workplace to investigate the accuracy and comfort of the off-the-shelf wearable sensor devices to monitor body temperature. Furthermore, it stands as the inaugural study in construction aimed at measuring temperature in the most common sites of the body as recommended by clinical guidelines [31], including the inner ear and axilla, using the off-the-shelf devices. The study aims to investigate the reliability, comfort, and accuracy of these devices in measuring workers' body temperature. The outcome of this research will aid in assessing the feasibility of adopting off-the-shelf body temperature wearable sensor devices in real construction sites to detect significant safety and health risks such as physical fatigue, heat stress and infectious deceases in an objective way.

2. Methodology

A prospective feasibility study design was used to investigate the temperature measurement accuracy using wearable sensors and the comfort of the devices. Construction workers from a range of diverse occupations were recruited to participate in this research. The

Table 1
Demographic information of the participants.

Construction site	Trade	Number of participants	Age (years) Minimu-maximum	Gender (M, F)
Site 1	Electrician	2	(23,32)	2,0
	Labourer	1	52	1,0
	OHS Representative	1	59	1,0
	Manager	1	27	1,0
	Plumber	5	(24, 66)	5,1
Site 2	Carpenter	1	39	1,0
	Manager	4	(23, 34)	3,1
	Labourer	5	(33, 57)	5,0
	Electrician	3	(25, 34)	3,0
	Plumber	1	40	1,0
	Supervisor	2	(25, 28)	2,0

workers were fitted with selected wearable sensors to monitor their body temperature during working hours continuously. A custom-built phone application was used to transfer the temperature readings from the wearable sensors to an advanced temperature monitoring dashboard in real time. The following presents the details of the methods used in this study.

2.1. Participants

Construction workers from two large construction sites in Geelong and Melbourne, Australia, were recruited for this research. The data collection was conducted from 10 (1 female) and 16 (1 female) volunteered participants in Geelong site (#1) and Melbourne site (#2), respectively. The data was collected during the working hours from May 03, 2021 to May 07, 2021 and June 22, 2021 to June 30, 2021 for site 1 and site 2, respectively. The study was conducted during the first variant of COVID-19 when body temperature was the most significant symptom of COVID. Temperature monitoring was also selected as the focus of the current study since it is one of the most important indicators of other significant safety risks on construction sites such as heat stress and physical fatigue. The participants were engaged in a range of occupations from office-based jobs to plumbers and electricians. Table 1 presents the demographic information of the participants.

2.2. Selection of the sensor devices

The off-the-shelf wearable sensor devices on the market with the ability to measure body temperature were explored and 9 sensor devices were initially shortlisted (Table 2). Before commencing the actual data collection in real construction sites, the feasibility of using these sensors for the current study was further investigated considering three main criteria including (1) access to raw data of the sensors (2) ability for continuous monitoring of the temperature and (3) the site of the body which is measured by the sensor. Access to raw data was needed to transfer temperature measurements to the central monitoring dashboard for future applications. This can include assessing the level of significant risks such as physical fatigue and heat stress. However, access to raw data was offered by only 4 suppliers out of 9. All the sensors except one met the requirement to report temperature values in near real-time. Finally, only three sensors out of 9 were able to measure the temperature of a site body which is recommended by reference clinical guides such as RACGP [31]. It should be stated that Axilla, inner ear (tympanic), and oral are the recommended sites of the body that give accurate body temperatures [37–39]. However, among those three sites of the body, it is not possible to measure oral temperature continuously.

The research team conducted several rounds of trial data collection in the laboratory to consider all the mentioned selection criteria. Finally, three off-the-shelf wearable sensor devices were selected including one tympanic sensor (Cosinuss) and two axillary sensors (VivaLNK and Equivital). These sensors measure the temperature of either axilla or tympanic as the two recommended sites of the body. These sensors are shown in Figs. 1 and 2 (a-c). The following section provides more details about the three selected sensors.

2.2.1. VivaLNK

VivaLNK (VivaLNK, California, USA) is the axillary body temperature monitoring sensor used in this research (Figs. 1(a) & 2(a)). This sensor is a medical device for real-time body temperature monitoring. The device is reusable, rechargeable, and is attached to the underarm using adhesives. This device has been previously used at the clinical level to monitor body temperature for different purposes [40,41].

2.2.2. Cosinuss

The ear sensor used in the study was °Temp- Cosinuss° (Cosinuss° GmbH, München, Germany) (Figs. 1(b) & 2(b)). Cosinuss is a real-time body temperature monitoring sensor that measures body temperature through the inner ear. This temperature screening tool can potentially be used for physically demanding work [42,43].

2.2.3. Equivital

The EQ02 (Equivital EQ02, Hidalgo, UK) is another device used to measure body temperature (Figs. 1(c) & 2(c)). The EQ02 is a reliable device for monitoring several physiological parameters such as body temperature, ECG, electrodermal activity [44]. Only body temperature monitoring was used for the purposes of this study. The EQ02 consists of a sensor electronic module (SME) and a belt of varying sizes. The participants select the comfortable belt size, and the belt holds the SME connected to their body to monitor their

Table 2
List of the identified off-the-shelf wearable sensor devices with ability to measure body temperature.

Device name	Measurement site	Access to raw data	Real time data	Selected (Yes/No)
Kenzen	Upper arm	No	Yes	No
Oura	Finger	No	No	No
Empatica	Wrist	Yes	Yes	No
Fitbit	Wrist	No	Yes	No
iFever	Wrist	No	Yes	No
Tcore	Forehead	No	Yes	No
Equivital	Axilla	Yes	Yes	Yes
VivaLNK	Axilla	Yes	Yes	Yes
Cosinuss	Inner ear (tympanic)	Yes	Yes	Yes



Fig. 1. Wearable temperature monitoring devices used in this study a) VivaLNK b) °Temp-Cosinuss° c) Equivital EQ02.

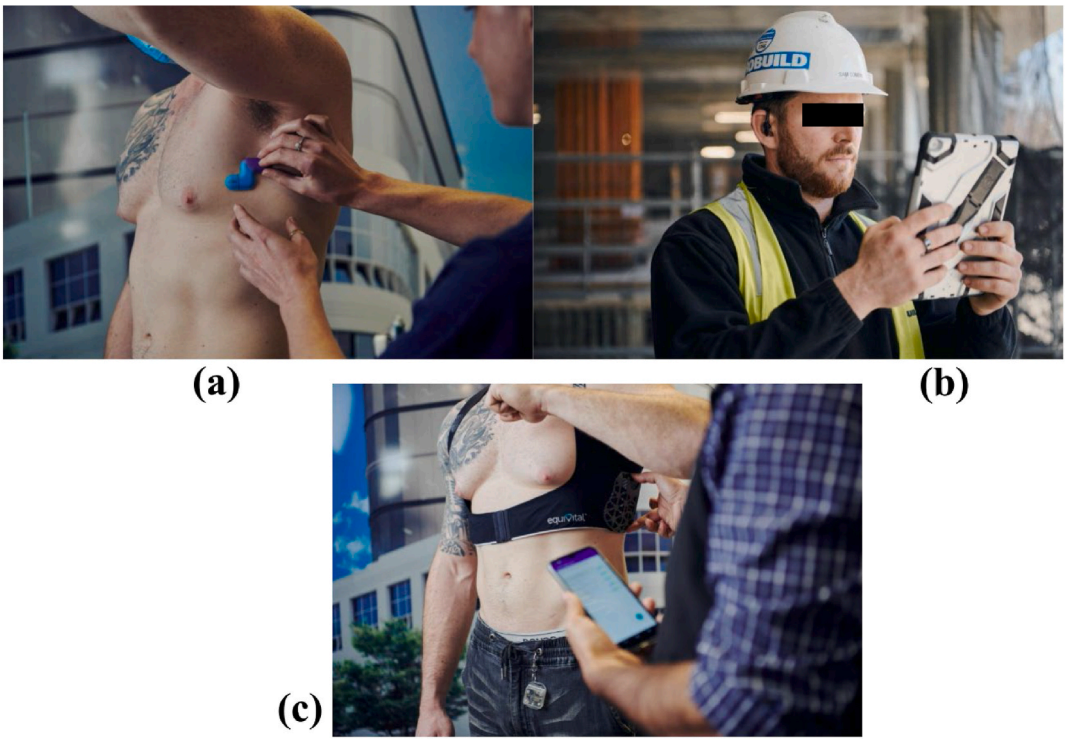


Fig. 2. Data collection in Melbourne and Geelong construction sites using the three selected wearable sensors a) VivaLNK b) °Temp- Cosinuss° c) Equivital EQ02.

body temperature.

2.2.4. *Reference devices*

A hospital-grade digital ear thermometer (Braun Thermoscan 7 IRT6520 Thermometer) and digital axilla thermometer (Surgipack Digital Thermometer 6344, Vega Technologies, China) was used to validate the temperature readings from the wearable sensors. These devices are commercially available for hospital use, and they are regarded as a benchmark for temperature monitoring accuracy [43, 45]. To receive the most accurate reference temperature readings from this device, all the measurements were conducted by an experienced registered nurse. All the reference temperature measurements were conducted in a room with a constant Ta (ambient temperature) of 20 ± 2°Ċ and RH (relative humidity) of 40 ± 10%. Fig. 3 shows the reference temperature reading from one of the workers using the ear thermometer.

2.3. *Data transferring tools*

In this project, a custom-built advanced temperature monitoring dashboard was developed to show temperature readings and ambient conditions in real time. Android smartphones (Motorola E7 Power) were employed to collect the data from the sensors



Fig. 3. Reference temperature reading on construction site by the registered nurse using the ear medical-grade thermometer.

attached to the participants' bodies during the working day. During data collection, workers carried a smartphone with a custom-built application to connect, read, and administer the sensors and transfer readings to the cloud and the monitoring dashboard. The readings from the three wearable sensors were sent to the mobile phone through Bluetooth and then the phone uploaded the data on the cloud. The participants were told to keep the mobile phone close to their bodies during the day to avoid disconnection between the phone and the sensors.

2.4. Data collection procedure

All the 26 participants were asked to wear three selected commercially available wearable sensors to measure body temperature during the working hours from 7 a.m. to around 4 p.m. Upon arrival at the data collection site in the morning, the participants were fitted with the three devices. The wearable sensors were connected to the mobile phones to collect and transfer data to the monitoring dashboard. The accuracy of measured temperatures was validated using the reference devices at three periods during the day including before the start of the work (pre-work), during the rest (smoking time and lunchtime), and after finishing the work (post-work). In each of these periods, they were asked to stay in the data collection site for 20 min and their body temperature was measured using the reference devices every 2 min (Fig. 4). In addition, the participants were asked to attend the data collection site and have a reference temperature reading whenever they passed by the data collection site during the day. The ambient condition in the data collection site was constant ($20 \pm 2^{\circ}\dot{C}$ for ambient temperature and $40 \pm 10\%$ for relative humidity). This was based on the expectation that the most reliable results would be achieved during the rest, pre-work, and post-work periods when the effect of ambient conditions (radiation, weather temperature, wind, and humidity) on body temperature fluctuations is minimised [46].

In this project, an advanced temperature monitoring dashboard was developed to show temperature readings and ambient conditions in real time. During data collection, workers carried an Android smartphone with a custom-built application to connect, read, and administer the sensors and transfer readings to the cloud and the monitoring dashboard. Fig. 5 shows the data acquisition process.

In order to collect the most reliable results, all details regarding the data collection procedure were examined in the laboratory by conducting many rounds of trial data collection before real data collection. In addition, researchers tried to simulate the working conditions in construction sites to identify any potential issue that might happen during real data collection.

2.5. Human factors evaluation

The workers who participated in this project completed a human factors questionnaire at the end of data collection. Each provided their opinions about the comfort and fit of the temperature sensor devices used during data collection to enable the research team to decide on the most comfortable device. The questionnaire asked specifically about (1) the participants' opinion about comfortability of

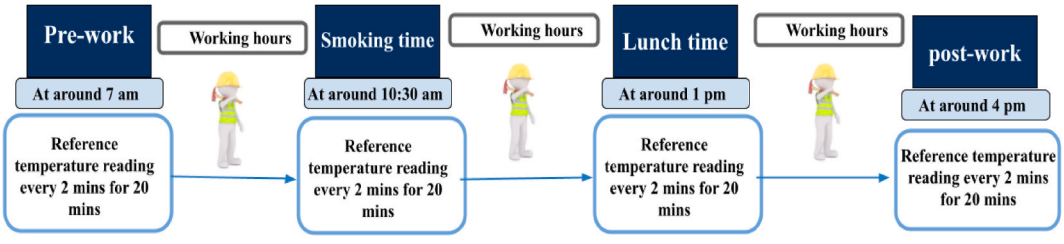


Fig. 4. Data collection procedure in each working day (The workers were fitted with the wearable sensors during working hours and four 20 min times, including pre-work, smoking time, lunch time and post-work were determined for reference temperature reading).

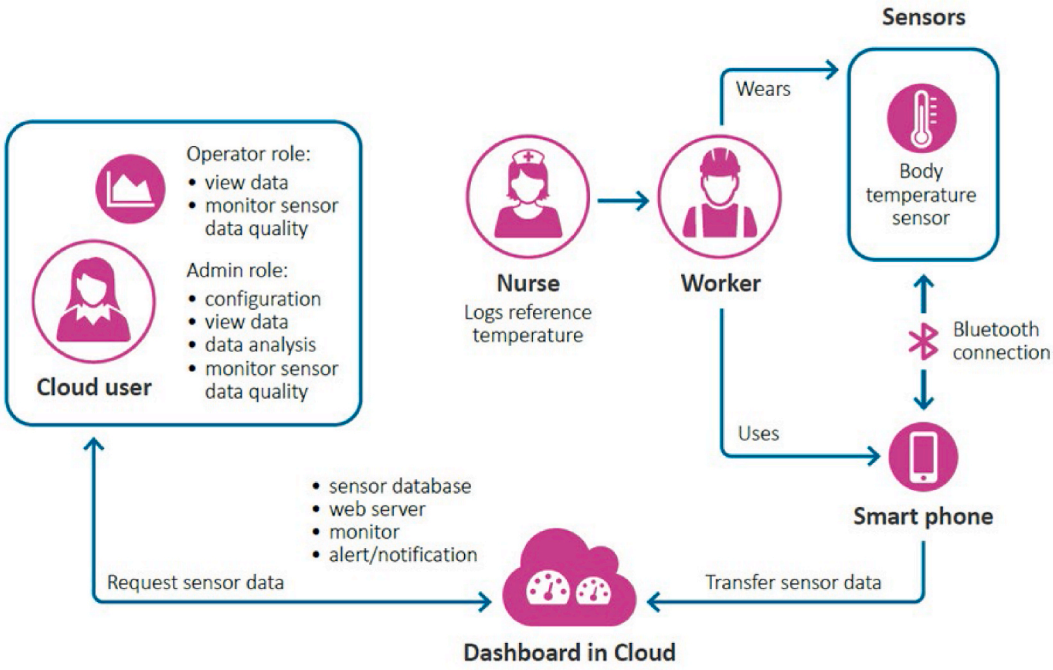


Fig. 5. Data acquisition process (Workers fitted with the wearable sensors and their body temperature were recorded as reference points during the day by a registered nurse. The sensors were connected to the cloud and the monitoring dashboard through a custom-built phone application. Data quality monitoring and data analysis were conducted by the cloudadmin).

the sensors, (2) the impact of the sensors on workers' performance, (3) the acceptability of the sensors to be worn for an extended period of 8 h or more and (4) finally the participants were asked to indicate the overall rank of the sensors.

2.6. Bland-Altman analysis

The focus of this study was to measure the superiority of temperature sensors in measuring the body temperature of construction workers. Such measurements may always imply some degree of error which needs to be assessed. Correlation may not be a suitable measure to account for this error, since correlation studies the relationship between one variable and another, but not the difference between them. The literature suggests the use of Bland-Altman plots which are based on the quantification of the agreement between two quantitative measurements. This is done by studying the mean difference and constructing limits of agreement [47]. The limits of agreement (LoA) are calculated as ± 1.96*SD difference and the assumptions of normality of difference are understood by using a graphical approach (i.e., Bland-Altman plot).

In this study, the difference in temperature readings (per minute) of three sensors from the two reference thermometers (ear and axillary thermometers) readings was studied. This produced two sets of temperature differences – one set for each reference thermometer. The difference between two paired temperature readings (one reading from a sensor and the other from a reference thermometer) is plotted against the mean of the two temperatures as a scatter plot XY, in which the Y axis shows the difference, and the X axis represents the average temperature. Bland-Altman plot recommends that 95% of the data points should lie within ± 1.96SD of the mean difference. The bias in the measurement is shown as a mean difference and being a positive value indicates that the first variable is measuring lower than the second variable and vice versa. The precision of estimated limits of agreement can be visualized through a 95% confidence interval (CI) of mean difference which illustrates the magnitude of the systematic difference. This can be understood by the relative location of the line of equality. For example, if the line of equality is not in the interval, there is a significant systematic difference, meaning that the second method is constantly under- or over-estimating compared to the first method.

3. Results

3.1. Comparative performance of sensors in measuring body temperature

Table 3 shows the mean temperature difference (MD) of each body temperature sensor (VivaLNK, Cosinuss, and Equivital) with respect to two reference thermometers - hospital grade ear (Ref 1) and axillary thermometers (Ref 2) for data collected over three weeks in two construction sites.

For the mean difference and variance of body temperatures w.r.t. the reference ear thermometer (Ref 1) during the 5 days of data

collection in site 1, the Cosinuss sensor shows the lowest for 4 out of 5 days (0.2, 0.2, 0.3, and 0.2 on 3rd and 5-7th of 2021/05), and the VivaLNK shows the lowest for the remaining day (0.2 on 2021/05/04). While in site 2, the VivaLNK readings are the lowest for 4 out of 6 days (0, 0.2, 0.5, and 0 for the 22-23rd, 25th, and 30th of 2021/06), the Equivital is the lowest for the remaining 2 days (0.4 and 0.2 for 24th and 29th on 2021/06). The Cosinuss reading, however, was close to the minimum for a single day (0.1 on 2021/06/30). The average mean difference for all three temperature sensors w.r.t. the reference ear thermometer is mostly above 0.37 °C.

With respect to the reference axillary thermometer (Ref 2), the mean difference and variance of body temperatures during the 5 days data collection in site 1, the Cosinuss shows the lowest values compared to other sensors for 3 out of 5 days (2.0, 0.7 and −0.2 on 4-5th, and 7th of 2021/05), the Equivital is the lowest for 2 remaining days (0 and −0.1 on 3rd and 6th of 2021/05). While in site 2, the Cosinuss is the lowest for 3 out of 6 days (0.6, 0.7, and −0.3 on 22nd, 24th, and 29th of 2021/06), the VivaLNK is the lowest for 3 days (0.6, 0.2, and −0.3 on 22-23rd and 25th of 2021/06, overlapping on 2021/06/22 with Cosinuss), and the Equivital is the lowest for the remaining one day (−0.4 on 2021/06/30).

In contrast to the reference ear thermometer, the mean differences w.r.t. the reference axillary thermometer are smaller and often show negative magnitudes. This is probably due to the differences between the two reference measures, which can be justified by the fact that the temperature of the inner ear is relatively higher than the axilla based on reference clinical guides [31].

Fig. 6(a–c) shows a Bland-Altman plot for three body-temperature sensors including Cosinuss, VivaLNK, and Equivital sensors with reference to a hospital-grade inner-ear thermometer. The bias of all three sensors is positive and the VivaLNK sensor readings show the lowest mean difference 0.40 followed by the Cosinuss and Equivital sensor readings of 0.45 and 0.67 in increasing order. The line of equality (zero mark in Y axis) for all three sensors is not in the mean difference's 95% confidence interval (CI) (blue shaded line in Fig. 6 a-c). The VivaLNK sensor readings also show the smallest distance 2.86 between the lower and upper bounds, followed by the Cosinuss 3.12 and Equivital 4.12. The Cosinuss and Equivital plots (Fig. 6 (a,c)) show <1% and VivaLNK shows around 2% data points outside the lower and upper boundaries (each boundary is highlighted in orange).

Fig. 7(a–c), shows Bland-Altman plot for three body-temperature sensors with reference to a hospital grade axillary thermometer. The bias of all the sensors is negative and the Equivital sensor readings show the lowest mean difference −0.22 followed by the Cosinuss −0.31 and VivaLNK −0.62 in an increasing order. The Cosinuss and Equivital plots (Fig. 7 (a,b)) shows the line of equality within the mean difference's 95% CI, but that of the VivaLNK is outside the 95% CI interval (Fig. 7(b)). The smallest distance between the lower and upper bounds of the mean difference is for VivaLNK 3.2, followed by Equivital 3.67 and Cosinuss 3.98. Less than 1% of the data points of all the sensors are outside of the lower and upper boundaries (Fig. 7(a–c)).

Based on the above-mentioned results, it is obvious that the sensor devices had shown better agreement w.r.t. the axillary reference thermometer compared to the reference ear thermometer. The Cosinuss and Equivital sensors were showing better agreements w.r.t. the axillary reference thermometer since their equality line lies within 95% CI of the mean difference. In addition, VivaLNK also performed well since the distance between the upper and lower bounds is smaller.

The sensors were compared under isothermal conditions (the ambient temperature was controlled to have a steady ambient temperature at 24 °C) and the participants were in rest condition. The readings from the sensors were compared against the hospital-grade inner ear thermometer. The average difference between the reference thermometer and the standard deviation of the differences are shown in Table 4. The results show that there are some differences between the readings from the three sensors and the reference thermometer. These differences were expected because of the positioning of the devices on the participants' bodies. The temperature of the inner ear (Cosinuss) is relatively higher than the axilla (VivaLNK and Equivital) based on reference clinical guides [31]. Therefore, the differences do not represent the effect of uncertainty in the calibration process and there are no systematic differences in the results.

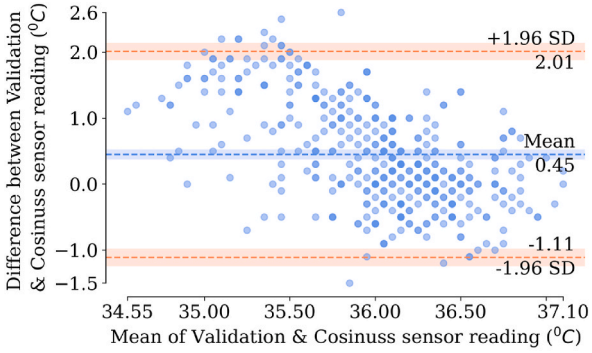
3.2. Human factors survey results

Fig. 8 shows human factor evaluation results based on the opinions of 26 workers from the two sites. The level of comfort

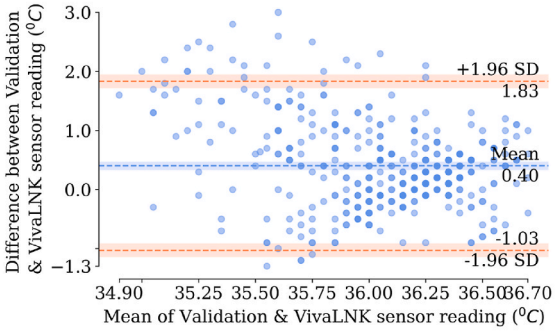
Table 3
Mean difference (MD) of body temperature (T) (°C) measurements in three sensor devices (VivaLNK, Cosinuss, and Equivital) versus Ref 1 (hospital grade ear thermometer) & Ref 2 (hospital grade axillary thermometer) for data collected over 11 days in 2 construction sites.

Site	Day	Temperature (T), MD±SD					
		VivaLNK vs Ref 1	Cosinuss vs Ref 1	Equivital vs Ref 1	VivaLNK vs Ref 2	Cosinuss vs Ref 2	Equivital vs Ref 2
Site 1	2021/05/03	0.5 ± 0.8	0.2 ± 0.7	0.8 ± 1.0	−0.8 ± 0.9	−1.6 ± 0.2	0.0 ± 0.7
	2021/05/04	0.2 ± 0.6	0.8 ± 0.8	0.9 ± 0.9	2.4 ± 0.1	2.0 ± 0.1	2.3 ± 0.1
	2021/05/05	0.5 ± 0.7	0.2 ± 0.7	1.2 ± 0.9	−0.9 ± 0.1	0.7 ± 0.3	3.4 ± 0.1
	2021/05/06	0.6 ± 0.7	0.3 ± 0.6	0.8 ± 1.1	−0.6 ± 1.0	−1.1 ± 0.8	−0.1 ± 1.1
	2021/05/07	0.7 ± 0.7	0.2 ± 0.7	0.6 ± 1.3	−0.8 ± 0.5	−0.2 ± 1.2	−0.7 ± 1.0
Site 2	2021/06/22	0.0 ± 0.6	1.4 ± 0.4	1.1 ± 0.8	0.6 ± 1.0	0.6 ± 0.8	2.7 ± 0.5
	2021/06/23	0.2 ± 0.7	0.5 ± 0.7	1.0 ± 1.0	0.2 ± 0.1	1.1 ± 0.1	0.8 ± 0.5
	2021/06/24	0.5 ± 0.9	1.1 ± 0.9	0.4 ± 1.0	2.7 ± 0.1	0.7 ± 0.7	1.2 ± 0.1
	2021/06/25	0.5 ± 0.5	0.7 ± 0.6	0.9 ± 0.8	−0.3 ± 0.5	−0.7 ± 0.2	−1.8 ± 0.1
	2021/06/29	0.3 ± 0.7	0.5 ± 0.7	0.2 ± 0.5	−0.9 ± 0.1	−0.3 ± 0.7	−0.3 ± 0.7
	2021/06/30	0.0 ± 0.6	0.1 ± 0.8	0.1 ± 1.2	−1.4 ± 0.4	−0.6 ± 0.7	−0.4 ± 1.1

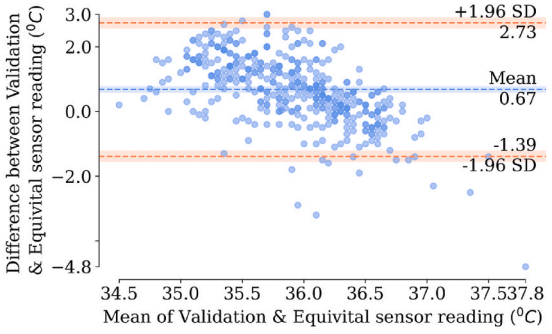
Ref 1: Hospital grade ear thermometer.
Ref 2: Hospital grade axillary thermometer.



(a)



(b)

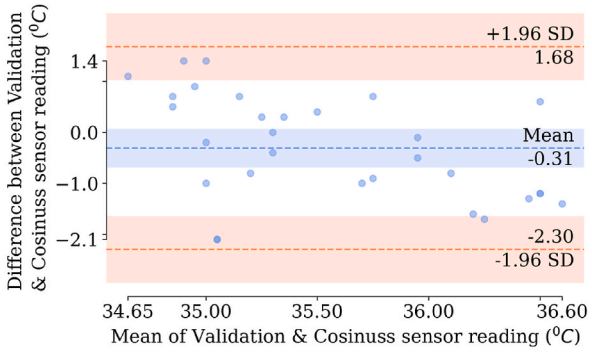


(c)

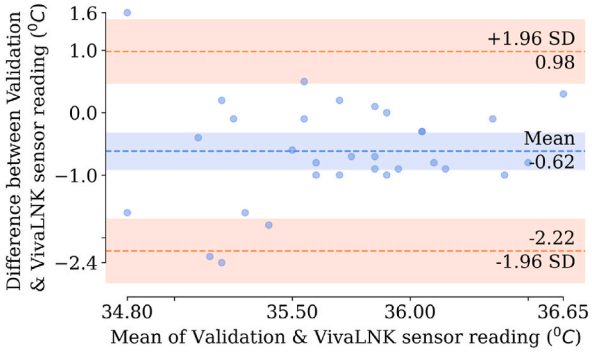
Fig. 6. Bland-Altman plot of three body-temperature sensors including (a) Cosinuss, (b) VivaLNK, and (c) Equivital sensor with reference to hospital grade inner ear thermometer.

experienced for each wearable sensor is shown in Fig. 8(a). The workers were asked to rate how comfortable they found each of the three wearable sensors during the test. A three-point scale was used to assess the level of comfort, with 1 being 'slightly-comfortable', 2 being 'moderately-comfortable' and 3 being 'very-comfortable'. As shown, 80% of workers believed the Equivital sensor is 'very-comfortable' or 'moderately-comfortable'. This shows that this sensor device has the potential to be used in construction sites without causing any discomfort. The VivaLNK sensor also showed acceptable comfort and achieved the second rank. However, only 36% of participants found the Cosinuss sensor to be 'very-comfortable' or 'moderately-comfortable'.

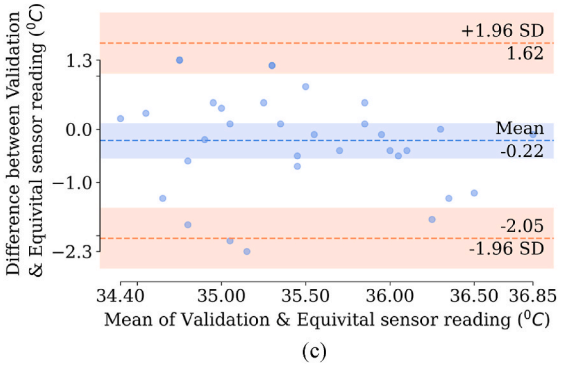
The impact of sensors on workers' performance is presented in Fig. 8(b). A 3-point scale was used to assess the impact, with 1 being 'moderate negative impact', 2 being 'slight negative impact', and 3 being 'no negative impact'. As shown in Figs. 6(b), 92% of workers believed that the Equivital and VivaLNK sensors have 'no negative impact' or 'slightly negative impact' on the performance of construction workers. This shows that these two sensor devices have the potential to be used in construction sites without causing any



(a)



(b)



(c)

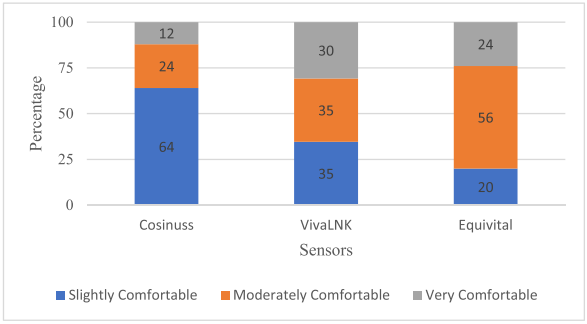
Fig. 7. Bland-Altman plot of three body-temperature sensors including (a) Cosinuss, (b) VivaLNK,and (c) Equivital sensor with reference to the hospital grade axillary thermometer.

Table 4
Comparison of the results for the readings from the sensors against the hospital grade inner ear thermometer under isothermal condition.

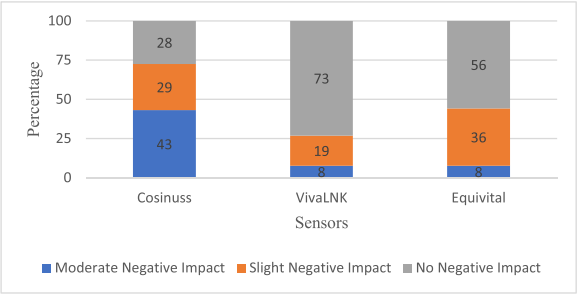
	Cosinuss	VivaLNK	Equivital
Average difference from the reference thermometer	0.2875	−0.225	−0.5375
Standard deviation of the differences	0.355097	0.549432	0.867378

negative impact on the overall performance of workers. However, only 56% of participants found the Cosinuss sensor has 'no negative impact' or 'slightly negative impact' on their performance.

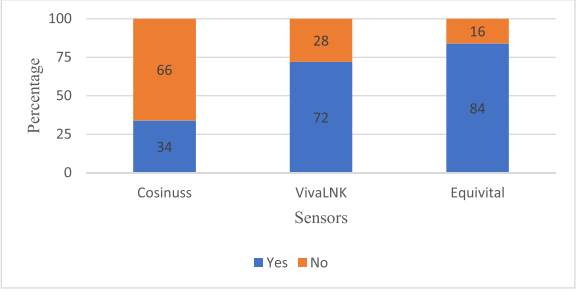
Finally, the workers were asked whether the three temperature sensor devices were acceptable to wear for an extended period of 8 h or more. Their answers are shown in Fig. 8(c). As shown, 73% of workers believed the Equivital sensor could be worn for an extended period of 8 h or more. This shows that this sensor device has the potential to be used in construction sites without causing any potential



(a)



(b)



(c)

Fig. 8. Human factors evaluation based on the opinions of 26 workers: (a) the level of comfort experienced for each of the 3 wearable sensors, (b) the impact of each wearable sensor on the overall performance of workers, and (c) the practicality of wearing sensor devices for an extended period of 8 h or more.

issues. This percentage was 62 and 31 for the VivaLNK and Cosinuss sensors, respectively.

4. Discussion

Three wearable sensors were used to measure the body temperatures of 26 construction workers from two sites over 11 days where two sensors were measuring body temperatures from the axilla (Equivital and VivaLNK sensor) and the other from the inner ear (Cosinuss sensor). Two hospital-grade ear and axillary thermometers were used as the reference devices in this study to evaluate the accuracy of the wearable sensors' temperature measurements.

The reference ear thermometer (Ref 1) measured comparatively higher body temperatures compared to the reference axillary thermometer (Ref 2) which can be observed from a higher positive mean temperature difference w.r.t. Ref 1 compared to Ref 2 (as shown in Table 3). The variation of measured temperature w.r.t reference ear thermometer was higher than the reference axillary thermometer, which can be attributed to the inherent differences between two reference measures. Based on reference clinical guides such as RACGP [31], the temperature of the inner ear is relatively higher than the axilla.

Although Cosinuss and reference ear thermometers both measure temperature from the inner ear, they use different technology in measuring body temperatures - Cosinuss makes contact inside the outer ear canal and the ear reference thermometer represents ear canal infrared thermometry [43].

With respect to the reference axillary thermometer, out of 11 days, the lowest mean difference was found for 6 days (days 2–3, and 5 in site 1, day 1, 3 and 5 in site 2), and 3 days (days 1–2 and 4 in site 2) for Cosinuss, and Vivalink, respectively (Table 3). The

equivital, on the other hand, shows the lowest mean difference for 2 days (day 5–6 in site 2) w.r.t. the ear and axillary reference thermometer (Table 3). This means, that although Cosinuss showed the lowest mean difference for a maximum number of days, Vivalink and Equivital sensors are also capable of measuring temperature with certain accuracy and relatively higher variations. In the future, further exploration is necessary to find the reason for performance variation and this needs to be addressed to improve the reliability of the measurements.

Very few (<1%) data points were found outside the lower and upper boundaries of Bland-Altman plots (Fig. 6) of three sensor devices with reference to hospital-grade inner ear thermometer. This indicates that all sensor devices measured temperatures with acceptable accuracy. Now if we consider the range of variation for each device (i.e. the difference between the upper and lower bounds) the VivaLNK sensor device shows the lowest range (2.86) compared to the other two devices - Cosinuss sensor device (3.12) and Equivital sensor device (4.12). This observation indicates the superiority of the VivaLNK sensor device among others in terms of temperature measuring performance w.r.t. the reference ear thermometer. In addition, the bias observed in the Bland-Altman plot (Fig. 6) with the reference ear thermometer for all three sensors is positive. The mean differences of all three sensors in the Bland-Altman plot were positive, which indicates the fact that the sensors were measuring body temperatures lower than the reference ear thermometer. In general, the relative position of the line of equality (the zero mark in the Y axis in Fig. 6) w.r.t. 95% CI of mean difference illustrates the magnitude of the systematic difference, while having the equality line within the 95% CI indicates insignificant reading or measurement difference. In this study (Fig. 6), all three sensors had the equality line outside the 95% CI which indicates that the sensors were underestimating the body temperature w.r.t. the reference ear thermometer.

When the hospital-grade axillary thermometer was used as the reference (Fig. 7), the range of variation (difference between upper and lower bounds) of VivaLNK sensor shows the lowest range (3.20) compared to the other two sensor devices Cosinuss sensor (3.98) and Equivital sensor (3.67). Compared to the VivaLNK sensor behaviour (lowest range of variation but the highest bias), the Equivital sensor seems better (lowest bias and 2nd lowest range of variation) with respect to the axillary reference temperature. In addition, the bias observed in the Bland-Altman plot with the reference axillary thermometer is negative for all the sensors (Fig. 7 shows the negative mean difference) which indicates that the sensors were measuring body temperatures higher than the reference thermometer. The line of equality of the VivaLNK sensor is outside the 95% CI which indicates that the VivaLNK sensor was overestimating compared to the reference axillary thermometer. On the other hand, the Cosinuss and Equivital sensors were found to be estimating reasonably since the line of equality was within the 95% CI for the reference axillary thermometer. This observation was in line with the above observation where Equivital was found to be a better choice since it was showing the lowest bias and small range of variation in the Bland-Altman plot (Fig. 7).

Part of the study was to evaluate human factors related to the usage and impact of these sensors which were investigated under 3 criteria including level of comfort, impact of sensors on performance, and their usability for a long period of time. The Equivital sensor was found to achieve the best ranking in all three criteria, followed by the VivaLNK. This preference can be justified by the fact that this sensor is used with a chest-belt which fits with the body conveniently and is not impacted by the body movement while working outdoors compared to the other two sensors. The VivaLNK was tied to the body with the help of a sticky patch which was subject to displacement with the body movement as was found during the data collection of this study. The Cosinuss received the lowest ranking possibly due to its location where it is worn in the ear which was required to be firmly tied into the ear hole to have better temperature readings.

In summary, the study aimed to test the feasibility of using wearable temperature sensors for continuous monitoring of body temperature in construction sites. The results of the study show that all three sensors including the Cosinuss, VivaLNK, and Equivital were measuring body temperature where the bias is < 1 and their level of agreement with regard to the axillary reference thermometer was better than the ear reference thermometer and they were found to show superiority with a relative decreasing order (the Cosinuss being the highest and Equivital being the lowest). However, in terms of comfort, usability, and their effect on worker performance, the Equivital was found to be the best followed by the VivaLNK and Cosinuss.

The outcomes of this research will enable researchers to analyze the accuracy and comfort of off-the-shelf wearable sensors and select the most appropriate device for continuous monitoring of body temperature with the aim of objective detection of health and safety risks. The developed body temperature monitoring platform can be used to design and develop objective assessment tools for the most significant health and safety risks such as heat stress and physical fatigue.

5. Limitations of the study and directions for future research

Wearable temperature sensors as one of the prime physiological biomarkers are foreseen to revolutionize the management of health and safety risks. Although this study is a step forward toward the practical application of wearable temperature sensors for objective monitoring of significant health and safety risks, there are some limitations that can be addressed in future research.

First, the current study focused on the accuracy and comfort of the wearable sensors to select the most appropriate device for continuous monitoring of body temperature. Future research can also analyze the cost of wearable devices by combining their short-term (capital) cost and long-term (operation/maintenance) cost. Second, we tried to conduct a comprehensive search to include the off-the-shelf wearable temperature sensors on the market, at the time the study was conducted in 2020–2021. Considering the rapid advancements in manufacturing wearable sensors, future research can adopt a similar approach to include the new off-the-shelf wearable temperature sensors and select the most appropriate device for continuous monitoring of body temperature. Lastly, future research can focus on advancements in manufacturing processes to address challenges like stability, repeatability, reliability, sensitivity, linearity, aging, and large-scale manufacturing, offering a future outlook for wearable systems [25,26].

6. Conclusion

Wearable sensors provide an objective tool for continuous monitoring of health and safety risks. Among various physiological parameters collected by wearable sensors, body temperature is the best biomarker for real-time monitoring of significant safety risks such as heat stress and physical fatigue as well as infectious deceases such as COVID. Despite this opportunity, there is a gap in the literature to investigate the potential use of off-the-shelf body temperature sensor devices in real construction sites considering their accuracy and comfort.

This project tested the feasibility of using off-the-shelf wearable temperature sensor devices in real-world work settings through an extensive data collection from two large construction sites located in Australia. Three wearable sensor devices, including two axillary (Equivital, and VivaLNK) and one ear-mounted (Cosinuss), were tested on 26 workers over three weeks. Two hospital-grade ear and axilla thermometers were used to validate the temperature readings The results showed that all three devices were measuring body temperature with a bias of less than one. The level of agreement of the three devices with regard to the axillary reference thermometer was better than the ear reference thermometer. Cosinus, VivaLNK, and Equivital exhibited varying levels of superiority in a relatively decreasing order based on measurement accuracy. However, in terms of comfort, acceptability to be worn for an extended period (>8hrs), and their effect on worker performance, Equivital was found to be the best, followed by VivaLNK and Cosinuss.

The project results indicate that using the selected off-the-shelf wearable sensors, it is feasible to develop a continuous and accurate body temperature monitoring framework for construction sites. This will be a step forward towards an objective monitoring of significant health and safety risks.

Ethics statement

This project has received the ethics approval (SEBE-2020-65) from the Deakin University Human Research Ethics Committee (HREC). The conducted study complied with all regulations and informed consents were obtained.

Data availability

The datasets analysed during the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Farnad Nasirzadeh: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chandan Karmakar:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Data curation. **Ahsan Habib:** Writing – original draft, Visualization, Formal analysis, Data curation. **Kevin Benny Neelangal:** Writing – original draft, Visualization, Investigation, Formal analysis. **Mostafa Mir:** Writing – original draft, Visualization, Investigation, Formal analysis. **SangHyun Lee:** Writing – review & editing, Methodology, Conceptualization. **Tony Arnel:** Writing – review & editing, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Farnad Nasirzadeh reports financial support was provided by Incolink. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.heliyon.2024.e26947.

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